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LINEAR ALGEBRA AND ITS APPLICATIONS

Kronecker product and SVD approximations in image restoration

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Abstract

Image restoration applications often result in ill-posed least squares problems involving large, structured matrices. One approach used extensively is to restore the image in the frequency domain, thus providing fast algorithms using fast Fourier transforms (FFTs). This is equivalent to using a circulant approximation to a given matrix. Iterative methods may also be used effectively by exploiting the structure of the matrix. While iterative schemes are more expensive than FFT-based methods, it has been demonstrated that they are capable of providing better restorations. As an alternative, we propose an approximate singular value decomposition (SVD), which can be used in a variety of applications. Used as a direct method, the computed restorations are comparable to iterative methods but are computationally less expensive. In addition, the approximate SVD may be used with the generalized cross validation method to choose regularization parameters. It is also demonstrated that the approximate SVD can be an effective preconditioner for iterative methods. © 1998 Elsevier Science Inc. All rights reserved.

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1. Introduction

In many engineering applications, matrix approximations are used to simplify and/or improve the efficiency of numerical methods. For example, when the conjugate gradient algorithm (CG) is used to solve a linear system of equations, an approximation to the coefficient matrix (called a preconditioner) can be used to speed convergence; see, for example, [15]. In signal and image processing, the need for fast algorithms often results in solving problems "in the Fourier domain". Typically, this is equivalent to using a circulant approximation to the coefficient matrix (the Fourier vectors are eigenvectors of all circulant matrices, cf. [12]). Circulant approximations are used extensively in image restoration [1,4,16,24].

Image restoration is the process of removing or minimizing degradations (typically blur) in an observed image. A linear discrete model of image restoration is the matrix-vector equation

$$\mathbf{g} = K\mathbf{f} + \mathbf{n}, \tag{1}$$

where $\mathbf{g}$ is the observed image, $\mathbf{f}$ is the true, or ideal image, $\mathbf{n}$ is additive noise, and K is a matrix that represents the blurring phenomena. Image restoration methods attempt to construct an approximation to $\mathbf{f}$ given $\mathbf{g}$, K, and in some cases statistical information about the noise. Two aspects regarding the matrix K make this process non-trivial:

- If the observed image array has dimension $n \times n$, then $\mathbf{f}$, $\mathbf{g}$, and $\mathbf{n}$ are vectors of length n^2, and K is an $n^2 \times n^2$ matrix. Typical values of n are 256, 512, and 1024, so the dimensions of the matrix K can be extremely large.
- The continuous model of image restoration is a first kind integral equation, which is well known to be very sensitive to perturbations in the data [36]. In the discrete form, this means the matrix K is severely ill-conditioned, with singular values decaying to, and clustering at zero. Thus, standard techniques to solve $K\mathbf{f} = \mathbf{g}$ are likely to produce solutions that are highly corrupted with noise.

The first problem can be addressed by observing that K often has structure that can be easily exploited. In particular, K is often a banded block Toeplitz matrix with banded Toeplitz blocks (BTTB). It is well known that matrix-vector multiplication can be performed very efficiently with such matrices using fast Fourier transforms (FFTs) [9], and therefore, iterative methods can be used effectively. Moreover, circulant approximations to K can be used as preconditioners (see the recent survey paper by Chan and Ng [9] and the references therein), as well as to restore the image in the frequency domain.

The second problem of noise sensitivity in the computed solution is more difficult, and is usually addressed using some form of regularization, including the truncated singular value decomposition [22], Tikhonov regularization (i.e., damped least squares) [17], truncated iterative methods [19], as well as other

approaches [20]. In this work, we do not wish to advocate one regularization scheme over another. Instead, we focus on the structure of the problem and consider an alternative to circulant approximations to K. In particular, we consider approximations involving Kronecker products. A Kronecker product is defined as

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{bmatrix}.$$

Our motivation for choosing Kronecker products is that some theoretical blurring operators (e.g., Gaussian) are separable, and therefore K can be factored as a Kronecker product of two matrices of size $n \times n$. Efficient numerical methods for image restoration that exploit the structure of a Kronecker decomposition have been considered; see, for example, [3,6,13].

In cases where K is not separable, it is shown in Section 2 that K can be approximated by a Kronecker product, or by a sum of only a few Kronecker products. Using these decompositions, it is shown how to construct an approximate singular value decomposition (SVD). Section 3 demonstrates the effectiveness of this approximate SVD in a variety of applications, and in Section 4 we provide some concluding remarks.

2. Approximation techniques in image restoration

Recall that linear discrete image restoration is modeled by the matrix vector Eq. (1). If we assume that the blurring operator is spatially invariant, then the matrix K is BTTB. The BTTB matrix K can therefore be uniquely determined by a single column $\mathbf{k}$ which contains all of the non-zero values in K; that is, some central column. This vector $\mathbf{k}$ can be obtained by imaging a single point source (i.e., unit vector). Specifically, if the $n \times n$ array P represents the image of a point source, then $\mathbf{k} = \mathrm{vec}(P^{\mathrm{T}})$, where the vec operator transforms matrices into vectors by stacking the columns. The center (peak) of the point source image corresponds to the diagonal entry. We can therefore use the notation $K = \mathrm{toep2}(\mathbf{k})$ to describe the banded BTTB matrix K. For example, suppose that

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix},$$

$$\max_{i,j} p_{ij} = p_{22}.$$

Then $\mathbf{k} = \mathrm{vec}(P^{\mathrm{T}})$ is the fifth column of K and

$$K = \left[\begin{array}{ccc|ccc|ccc}
p_{22} & p_{21} & 0 & p_{12} & p_{11} & 0 & 0 & 0 & 0 \\
p_{23} & p_{22} & p_{21} & p_{13} & p_{12} & p_{11} & 0 & 0 & 0 \\
0 & p_{23} & p_{22} & 0 & p_{13} & p_{12} & 0 & 0 & 0 \\
\hline
p_{32} & p_{21} & 0 & p_{22} & p_{21} & 0 & p_{12} & p_{11} & 0 \\
p_{33} & p_{32} & p_{31} & p_{23} & p_{22} & p_{21} & p_{13} & p_{12} & p_{11} \\
0 & p_{33} & p_{32} & 0 & p_{23} & p_{22} & 0 & p_{13} & p_{12} \\
\hline
0 & 0 & 0 & p_{32} & p_{31} & 0 & p_{22} & p_{21} & 0 \\
0 & 0 & 0 & p_{33} & p_{32} & p_{31} & p_{23} & p_{22} & p_{21} \\
0 & 0 & 0 & 0 & p_{33} & P_{32} & 0 & P_{23} & P_{22}
\end{array}\right].$$

In a similar manner, we use the notation $X = \mathrm{toep}(\mathbf{x})$ to represent a banded point Toeplitz matrix X constructed from the vector $\mathbf{x}$, with the assumption that the location of the diagonal entry is known. For example, suppose that the second component of the vector $\mathbf{x} = [\, x_1\ x_2\ x_3\ x_4 \,]^{\mathrm{T}}$ corresponds to the diagonal element of the banded Toeplitz matrix X. Then

$$X = \mathrm{toep}(\mathbf{x}) = \begin{bmatrix} x_2 & x_1 & 0 & 0 \\ x_3 & x_2 & x_1 & 0 \\ x_4 & x_3 & x_2 & x_1 \\ 0 & x_4 & x_3 & x_2 \end{bmatrix}.$$

It is important to note that K is never formed explicitly. Efficient matrix-vector multiplications can be performed using FFTs by implicitly embedding K into a $2n \times 2n$ block circulant matrix with $2n \times 2n$ circulant blocks (BCCB). The resulting $4n^2 \times 4n^2$ BCCB matrix C is uniquely determined by its first column, thus requiring only the storage of the $4n^2 \times 1$ vector $\mathbf{c}_1$. More details on the structure of K and efficient implementation of matrix-vector multiplications can be found in [21].

Iterative methods, such as CG, can be used effectively for image restoration problems [21,27], by taking advantage of the fast matrix-vector multiplication. Another approach used extensively is to restore the image in the frequency domain [1,4,16,24]. A BTTB matrix K can be approximated by a matrix C which is BCCB [7,11]. The matrix C is diagonalized by the two-dimensional Fourier matrix $\mathscr{F}$,

$$C = \mathscr{F}^{*} \Lambda \mathscr{F},$$

where $\mathscr{F} = F \otimes F$ and F denotes a unitary one-dimensional Fourier transform matrix [12]. The eigenvalues in Λ are obtained by taking a two-dimensional FFT of the first column of C. Therefore, C need not be explicitly formed

and the least squares problem $C\mathbf{f} = \mathbf{g}$ can be solved using two-dimensional FFTs. The *optimal circulant preconditioner* introduced by Chan [10] produces a circulant approximation C which minimizes $\|C - K\|_F$ over all circulant matrices. This approximation is extended to the block case by Chan and Olkin [11] and produces a BCCB matrix C which satisfies

$$\min_C \|C - K\|_F = \min_\Lambda \|\Lambda - \mathscr{F} K \mathscr{F}^*\|_F,$$

over all diagonal matrices Λ.

As an alternative approach, we propose an approximation involving Kronecker products and the SVD. Instead of approximating the matrix K directly, we consider the SVD $K = U\Sigma V^{\mathrm{T}}$. In applications where K is very large, it is generally not feasible to calculate the SVD of K explicitly. If the point source image can be expressed as $P = \sigma \mathbf{u}\mathbf{v}^{\mathrm{T}}$ (i.e., has rank 1), then the matrix K is separable. Therefore, $K = A \otimes B$, where $A = \mathrm{toep}(\sqrt{\sigma}\mathbf{u})$ and $B = \mathrm{toep}(\sqrt{\sigma}\mathbf{v})$ [2,26,30,32]. Given the SVDs of A and B:

$$A = U_A \Sigma_A V_A^{\mathrm{T}},$$

$$B = U_B \Sigma_B V_B^{\mathrm{T}},$$

the SVD of K can be expressed as

$$\begin{aligned} K &= (U_A \Sigma_A V_A^{\mathrm{T}}) \otimes (U_B \Sigma_B V_B^{\mathrm{T}}) \\ &= (U_A \otimes U_B)(\Sigma_A \otimes \Sigma_B)(V_A \otimes V_B)^{\mathrm{T}} \\ &= U \Sigma V^{\mathrm{T}}. \end{aligned}$$

Regularized least squares problems associated with $\mathbf{g} = K\mathbf{f} + \mathbf{n}$ can then be solved using the SVDs of A and B, without ever forming K or its SVD directly.

If the matrix K is not separable, it can be approximated as a sum of Kronecker products. Van Loan and Pitsianis [33] propose a general technique for such an approximation where $\|(A \otimes B) - K\|_F^2$ is minimized, and Pitsianis [29] extends this work to approximations where $\|\sum(A_i \otimes B_i) - K\|_F^2$ is minimized. In the context of image restoration, similar approximations can be computed very efficiently; see Nagy [26] and Thirumalai [32]. If $\mathrm{rank}(P) = r$, then

$$P = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^{\mathrm{T}},$$

and

$$K = \sum_{i=1}^{r} A_i \otimes B_i,$$

$$A_i = \mathrm{toep}(\mathbf{a}_i),$$

$$B_i = \mathrm{toep}(\mathbf{b}_i),$$

where $\mathbf{a}_i = \sqrt{\sigma_i}\mathbf{u}_i$ and $\mathbf{b}_i = \sqrt{\sigma_i}\mathbf{v}_i$. For an $n^2 \times n^2$ matrix K, P will be at most $n \times n$, and, in some cases, can be much smaller than the dimensions of the observed image. Therefore, computing the SVD of P is relatively inexpensive when compared to the cost of solving systems with K. In addition, as illustrated in [26], a good approximation to P can often be obtained by using only a few terms corresponding to the largest singular values.

We now describe how to construct an SVD approximation to K. Given $k \leqslant r$ and

$$K \approx \sum_{i=1}^{k} A_i \otimes B_i, \tag{2}$$

then an SVD approximation of K can be constructed as:

$$\begin{aligned} K &\approx U\Sigma V^{\mathrm{T}}, \\ U &= U_A \otimes U_B, \\ V &= V_A \otimes V_B, \\ \Sigma &= \operatorname{diag}(U^{\mathrm{T}}KV) \\ &= \operatorname{diag}(U^{\mathrm{T}}(A_1 \otimes B_1 + A_2 \otimes B_2 + \cdots + A_k \otimes B_k)V), \end{aligned}$$

where $A_1 = U_A\Sigma_A V_A^{\mathrm{T}}$ and $B_1 = U_B\Sigma_B V_B^{\mathrm{T}}$, as before. Note that the number of terms k used to approximate P only affects the amount of work required to calculate Σ. If we choose $k = 1$, then it is shown in [33] that A_1 and B_1 minimize $\|A \otimes B - K\|_F$ over all rank-one approximations $A \otimes B$. For $k \geqslant 1$, $\Sigma = \operatorname{diag}(U^{\mathrm{T}}KV)$ clearly satisfies

$$\min_{\Sigma} \|\Sigma - U^{\mathrm{T}}KV\|_F = \min_{\Sigma} \|U\Sigma V^{\mathrm{T}} - K\|_F,$$

over all diagonal matrices Σ and therefore produces an optimal SVD approximation, given a fixed $U = U_A \otimes U_B$ and $V = V_A \otimes V_B$. This is analogous to the circulant approximation discussed earlier, which provides an optimal eigendecomposition given a fixed set of eigenvectors (i.e., the Fourier vectors).

3. Applications

In this section, we consider several applications of the SVD approximation. Throughout this section, we define the matrix $\tilde{K}$ as

$$\tilde{K} \equiv U\Sigma V^{\mathrm{T}} = (U_A \otimes U_B)\Sigma(V_A \otimes V_B)^{\mathrm{T}},$$

where $U\Sigma V^{\mathrm{T}}$ is the optimal SVD approximation of K described in Section 2. In each application considered, we compare the analogous schemes obtained by using the optimal BCCB approximation to K, and we refer to this as the FFT-based method. We begin with a brief description of the test problems used in our numerical experiments.

Test problems from astronomical imaging. The first example is used by astronomers to test and compare image restoration algorithms for Hubble Space Telescope (HST) images; see, for example the Newsletter of the STScI's Image Restoration Project [18], and [25]. This data is intended to simulate a star cluster image taken by the HST before the camera was fixed. Fig. 1 shows the true star cluster, and the image as would be given by the HST. The images are 128×128 and the point spread function P, also shown in Fig. 1, is 60×60. The second data set was developed at the US Air Force Phillips Laboratory, Lasers and Imaging Directorate, Kirtland Air Force Base, New Mexico, and has also been used in [27,28,31,34]. The image is a computer simulation of a field experiment showing a satellite as taken from a ground based telescope. The true and blurred images are 256×256 and are shown in Fig. 2. Similarly, the given point spread function is the 256×256 simulated image of a guide star, also shown in Fig. 2. The third example, shown in Fig. 3, is a 128×128 cropped version of the satellite data. This example is used to demonstrate results on an image which contains less of the bland background. All numerical tests were performed on a DEC Alpha Station $200^{4/233}$ using Matlab 4.2c.

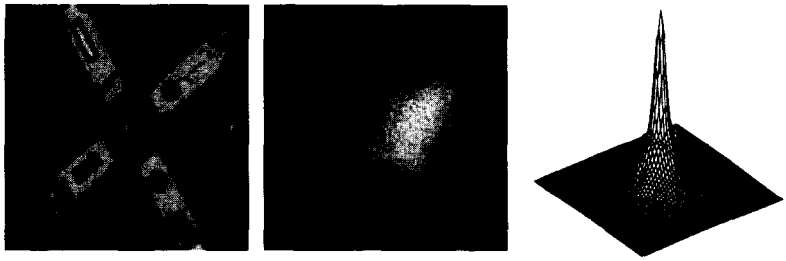
Tikhonov regularization and generalized cross validation. Since the image restoration problem is typically ill-posed, we are primarily interested in computing

a. Original image. b. HST simulated image. c. Point spread function.

Fig. 1. Original and HST simulated images.

a. Original image. b. Blurred image. c. Point spread function.

Fig. 2. Original and blurred satellite images.



a. Original subimage. b. Blurred subimage. c. Point spread function.

Fig. 3. Original and blurred satellite subimages.

a regularized solution. For this work, we use Tikhonov regularization to address the noise sensitivity of the computed solution. In Tikhonov regularization, a regularized solution $\mathbf{f}_{\text{reg}}$ is computed as the unique solution to [17]

$$\min_{\mathbf{f}}\{\|K\mathbf{f}-\mathbf{g}\|_2^2+\lambda^2\|\mathbf{f}\|_2^2\}.$$

The regularization parameter λ controls the "smoothness" of the regularized solution. This minimization problem can be reformulated as the damped least squares problem

$$\min_{\mathbf{f}}\left\|\begin{bmatrix}K\\ \lambda I\end{bmatrix}\mathbf{f}-\begin{bmatrix}\mathbf{g}\\ 0\end{bmatrix}\right\|_2^2. \tag{3}$$

The choice of the regularization parameter λ is a non-trivial issue, and much work has been done in this area; see, for example, the survey of regularization methods by Hanke and Hansen [20]. For this work, we choose one of the more common parameter choice methods, generalized cross-validation (GCV) [14,35]. The regularization parameter is chosen to minimize the GCV function

$$V(\lambda)=\frac{\|K\mathbf{f}_{\text{reg}}-\mathbf{g}\|_2^2}{[\operatorname{tr}(I-KK^I)]^2},$$

where K^I is the matrix which produces the regularized solution, i.e., $\mathbf{f}_{\text{reg}}=K^I\mathbf{g}$. For Tikhonov regularization, $K^I=(K^{\mathrm{T}}K+\lambda^2I)^{-1}K^{\mathrm{T}}$, which leads to

$$V(\lambda)=\frac{\|(I-T)\mathbf{g}\|_2^2}{[\operatorname{tr}(I-T)]^2}, \tag{4}$$

where $T=K(K^{\mathrm{T}}K+\lambda^2I)^{-1}K^{\mathrm{T}}$.

Application 1: *Estimating the regularization parameter*. An estimate of the regularization parameter can be found by evaluating the GCV function $V(\lambda)$ using the SVD of $\tilde{K}$. In particular, replacing K with $\tilde{K}$ in Eq. (4) and simplifying, we obtain

$$V(\lambda) = \frac{\sum_{i=1}^{n^2} \left(\frac{\hat{g}_i}{\sigma_i^2 + \lambda^2} \right)^2}{\sum_{i=1}^{n^2} \left(\frac{1}{\sigma_i^2 + \lambda^2} \right)^2}, \tag{5}$$

where $\hat{\mathbf{g}} = U^{\mathrm{T}}\mathbf{g}$. We then find the value of λ that minimizes Eq. (5). We note that a similar approach using the BCCB approximation, $C = \mathscr{F}^* \Lambda \mathscr{F}$ can easily be derived. In addition to using these estimates for the regularization parameter, we also compare results obtained using the best λ found by experimentation.

Application 2: *Use $\tilde{K}$ in the damped least squares problem.* The large dimensions of K suggests that direct factorization schemes may not be computationally feasible. However, if we use $\tilde{K}$ in place of K, the damped least squares problem (3) can be simplified to

$$\min \left\{ \begin{bmatrix} \Sigma \\ \lambda I \end{bmatrix} \mathbf{y} - \begin{bmatrix} \mathbf{z} \\ 0 \end{bmatrix} \right\},$$

where $\mathbf{y} = V^{\mathrm{T}}\mathbf{f}$ and $\mathbf{z} = U^{\mathrm{T}}\mathbf{g}$. This least squares problem is then trivial to solve. Note that, for $n \times n$ images and using $\tilde{K}$ in place of K, only $\mathrm{O}(n^3)$ arithmetic operations are needed to solve this Tikhonov regularized least squares problem.

A similar approach results when using an FFT-based method. That is, K is replaced by the optimal circulant approximation $C = \mathscr{F}^* \Lambda \mathscr{F}$, and two-dimensional FFTs are used to form matrix-vector multiplications with $\mathscr{F}$ and $\mathscr{F}^*$. This approach requires $\mathrm{O}(n^2 \log n)$ operations.

Application 3: *Use iterative methods, with $\tilde{K}$ as a preconditioner.* Finally we consider solving the damped least squares problem (3), with the BTTB matrix K, using a variant of the conjugate gradient method known as PCGLS; see [5] for a description and analysis of this algorithm. The approximate SVD is used in two contexts:

- to estimate the regularization parameter λ, as described above, and
- to construct a preconditioner.

The preconditioner can be constructed as follows. First note that

$$K^{\mathrm{T}}K + \lambda^2 I \approx (V_A \otimes V_B)(\Sigma^{\mathrm{T}}\Sigma + \lambda^2 I)(V_A \otimes V_B)^{\mathrm{T}},$$

so a preconditioner P can be taken to be

$$P = (V_A \otimes V_B)(\Sigma^{\mathrm{T}}\Sigma + \lambda^2 I)^{1/2}(V_A \otimes V_B)^{\mathrm{T}}.$$

This approach is similar to the one used in [8] for constructing a BCCB preconditioner. In our numerical tests, summarized below, we use no preconditioning, as well as $\tilde{K}$ and C to construct preconditioners. The convergence tolerance used in the algorithm is the relative residual: $\|\mathbf{r}_k\|_2 / \|\mathbf{g}\|_2 \leqslant 10^{-12}$, where $\mathbf{r}_k$ is the computed residual at the kth iteration.

Summary of results. As previously mentioned, two regularization parameters are used in each test. In one case, λ is selected using the GCV function, as described above. In the other case, λ is chosen experimentally by attempting to minimize the relative error (there are an infinite number of possibilities, so the optimal λ with respect to the relative error would be difficult to find exactly). While the experimental value of λ typically provided a better restoration, it is not a practical approach for realistic problems. Table 1 summarizes the estimated values of λ using these two approaches. Also shown in this table are the relative errors of the corresponding computed solutions (the PCGLS results were essentially the same regardless of the preconditioner). Corresponding computed restorations for these problems are shown in Fig. 4 (where λ is chosen by experimentation) and Fig. 5 (where the GCV function is used to estimate λ).

We note that for the satellite images, a good estimate of λ was obtained by using $\tilde{K}$ in the GCV function $V(\lambda)$. Although the results for the star cluster are not impressive, the computed restorations shown in Fig. 5 are comparable to the other schemes. The difficulty in locating the minimum in this case is probably due to the fact that the star cluster example is not very ill-conditioned; $V(\lambda)$ is fairly flat near the (very small) optimal λ. Similar behavior has been observed by Hansen [23]. The FFT-based method had difficulty with the satellite subimage, which we attribute to the fact that the true image is not approximated well by a periodic image. In this case, the optimal λ selected for the FFT-based method provided a solution which is reasonable, but clearly not as good as the PCGLS or approximate SVD solutions. The value of λ selected by GCV resulted in an FFT-based solution which is not meaningful, since λ is approximately an order of magnitude too small.

In addition, we observe that by using a direct method with $\tilde{K}$, we were able to compute restorations that were nearly as good as those obtained from PCGLS. In addition, combining this direct method with the GCV scheme for estimating λ, we obtain an $O(n^3)$ "black-box" approach that computes relatively good restorations. PCGLS is, at best, an $O(n^3 \log n)$ method [9].

Table 1
Relative errors

		Star cluster		Satellite		Satellite subimage	
		Exp.	GCV	Exp.	GCV	Exp.	GCV
FFT	λ	0.001	0.0213	0.0125	0.01	0.0225	0.0009
	Error	0.1383	0.1689	0.3614	0.3681	0.4259	6.955
SVD	λ	0.005	0.0314	0.0125	0.0096	0.015	0.0095
	Error	0.1393	0.2323	0.3466	0.3515	0.3629	0.3689
PCGLS	λ	0.001	0.0314	0.0125	0.0096	0.0125	0.0095
	Error	0.1343	0.2355	0.3484	0.3567	0.3606	0.3701

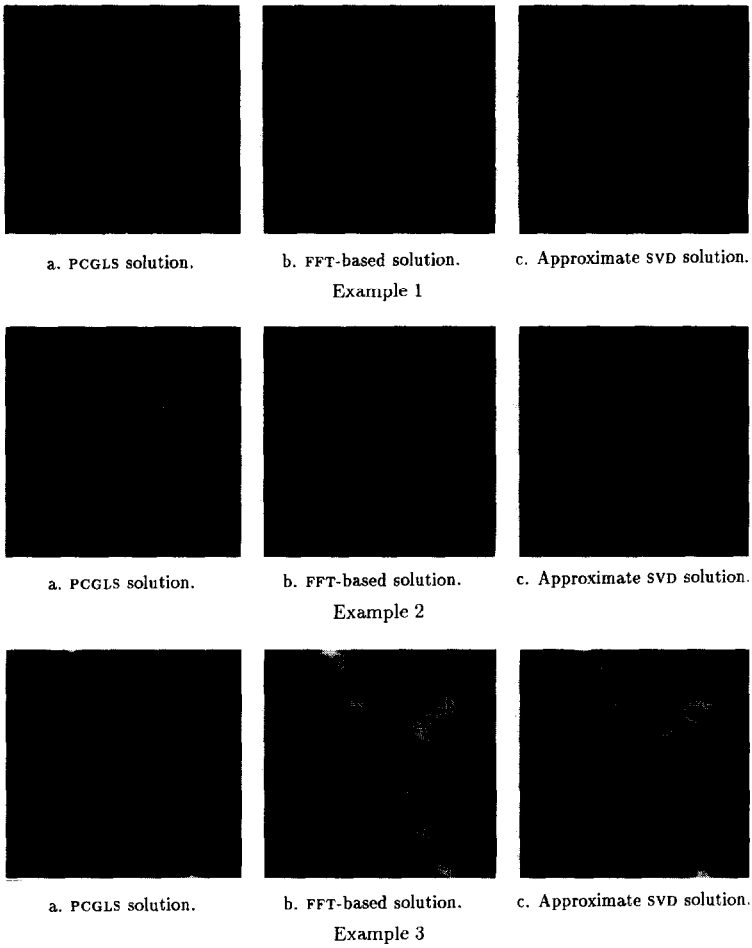
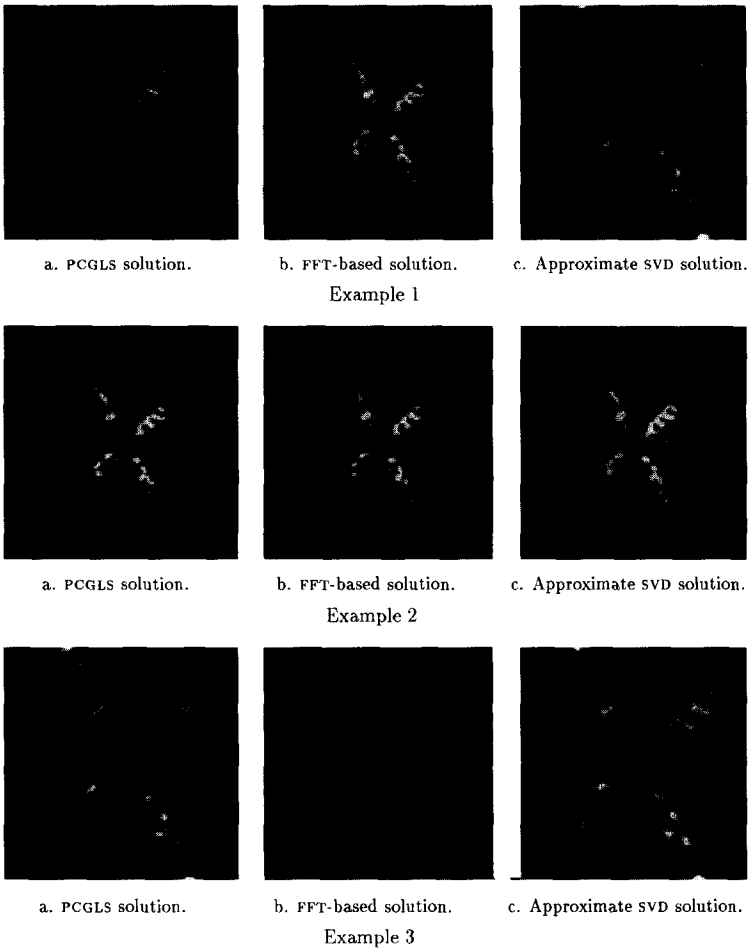


Fig. 4. Tikhonov regularization (λ selected experimentally).

Finally, we remark on the preconditioning aspects of $\tilde{K}$ and C. Table 2 shows the number of iterations for each problem, with each of the preconditioners. Due to the stringent stopping criteria (relative residual $\leqslant 10^{-12}$), these numbers are perhaps a bit deceiving. Therefore, in Fig. 6 we plot the relative errors for the first ten iterations. In the case of the star cluster, there is no apparent difference in the relative errors. That is, both $\tilde{K}$ and C are equally effective in reducing the number of iterations. However, in the satellite images, $\tilde{K}$ is much more effective than C. The initial increase in relative errors when using the BCCB preconditioner indicates that it is sensitive to noise, as well as to the value of λ. It should be noted, though, that applying the BCCB preconditioner at each iteration is only $O(n^2 \log n)$, while applying $\tilde{K}$ requires $O(n^3)$



a. PCGLS solution. b. FFT-based solution. c. Approximate SVD solution.

Example 1

a. PCGLS solution. b. FFT-based solution. c. Approximate SVD solution.

Example 2

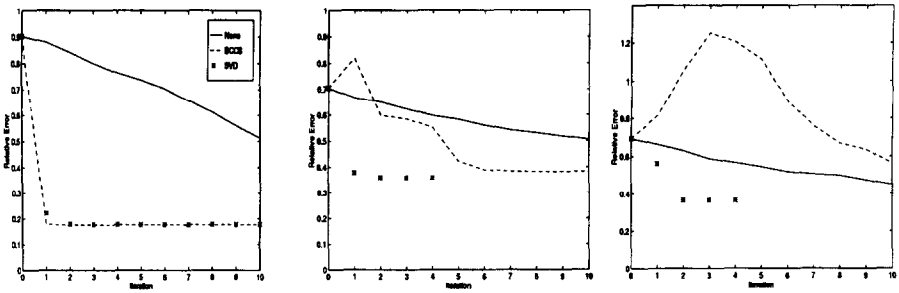
a. PCGLS solution. b. FFT-based solution. c. Approximate SVD solution.

Example 3

Fig. 5. Tikhonov regularization (λ selected by GCV).

Table 2
Number of PCGLS iterations

	Star cluster		Satellite		Satellite subimage	
	λ		λ		λ	
Preconditioner	0.001	0.0314	0.0125	0.0096	0.0125	0.0095
None	171	145	53	59	26	34
BCCB	34	32	30	34	34	37
Approx. SVD	18	15	4	4	4	5



a. Star cluster. b. Satellite. c. Satellite subimage.

Fig. 6. Relative errors at each iteration.

operations. Table 3 shows the actual amount of work (as measured by the Matlab function flops) for each test case, normalized by the total amount of work for no preconditioning. In some instances, it is not necessary to construct the preconditioner for each image. One may wish to experiment with several values of λ or may be restoring a number of images using the same blurring matrix. In these cases, the preconditioner can be constructed once and reused for each restoration. Table 4 shows the normalized amount of work for each test case, excluding the cost of constructing the preconditioner. In the case

Table 3
Relative FLOP count, including preconditioner setup

	Star cluster		Satellite		Satellite subimage	
	λ		λ		λ	
Preconditioner	0.001	0.0314	0.0125	0.0096	0.0125	0.0095
None	1.0	1.0	1.0	1.0	1.0	1.0
BCCB	0.2473	0.2782	0.7027	0.7025	1.54	1.4994
Approx. SVD	0.2870	0.2924	0.4798	0.4248	0.6118	0.6109

Table 4
Relative FLOP count, excluding preconditioner setup

	Star cluster		Satellite		Satellite subimage	
	λ		λ		λ	
Preconditioner	0.001	0.0314	0.0125	0.0096	0.0125	0.0095
None	1.0	1.0	1.0	1.0	1.0	1.0
BCCB	0.2470	0.2779	0.7018	0.7016	1.5381	1.4976
Approx. SVD	0.2615	0.2619	0.3084	0.2730	0.4214	0.4450

of the star cluster, both preconditioners were equally effective in reducing the total amount of work, whereas the approximate SVD preconditioner was clearly more efficient in the other test cases. However, execution time is greatly affected by the architecture and implementation used. The majority of the execution time is spent performing the necessary matrix multiplications and two-dimensional FFTs, for which we used the Matlab function fft2. It is well known that the execution time for the two-dimensional FFT varies greatly depending on the sequence length and whether the input sequence is real or complex. While our results using the approximate SVD are encouraging, the issues involved with developing the most efficient implementation have yet to be addressed.

4. Conclusions

In this paper, we have developed an approximate SVD that provides a viable alternative to FFT-based methods for solving large, structured least squares problems in image restoration. While direct FFT-based approaches are typically less expensive, the resulting solutions may be degraded due to the periodicity assumption implicit in such methods. The approximate SVD is demonstrated as a direct method providing performance similar to the iterative CG method, but at significantly less cost. It also provides a method for choosing regularization parameters, resulting in an efficient "black-box" approach to image restoration. Finally, we have shown that the approximate SVD can be very effective as a preconditioner for iterative methods.

References

[1] H. Andrews, B. Hunt, Digital Image Restoration, Prentice-Hall, Englewood Cliffs, NJ, 1977.
[2] H.C. Andrews, C.L. Patterson, Singular value decompositions and digital image processing, IEEE Trans. Acoust. Speech and Signal Proc. 24 (1976) 26–53.
[3] E.S. Angel, A.K. Jain, Restoration of images degraded by spatially varying pointspread functions by a conjugate gradient method, Applied Optics 17 (1978) 2186–2190.
[4] J. Biemond, R.L. Lagendijk, R.M. Mersereau, Iterative methods for image de-blurring, Proceedings IEEE 78 (1990) 856–883.
[5] Å. Björck, Numerical Methods for Least Squares Problems, SIAM, Philadelphia, PA, 1996.
[6] D. Calvetti, L. Reichel, Application of ADI iterative methods to the image restoration of noisy images, SIAM J. Matrix Anal. Appl. 17 (1996) 165–174.
[7] R.H. Chan, X.-Q. Jin, A family of block preconditioners for block systems, SIAM J. Sci. Stat. Comp. 13 (1992) 1218–1235.
[8] R.H. Chan, J.G. Nagy, R.J. Plemmons, FFT-based preconditioners for Toeplitz-block least squares, SIAM J. Numer. Anal. 30 (1993) 1740–1768.
[9] R.H. Chan, M.K. Ng, Conjugate gradient methods for Toeplitz systems, SIAM Review 38 (1996) 427–482.

[10] T.F. Chan, An optimal circulant preconditioner for Toeplitz systems, SIAM J. Sci. Stat. Comp. 9 (1988) 766–771.
[11] T.F. Chan, J.A. Olkin, Preconditioners for Toeplitz-block matrices, Numer. Algo. 6 (1993) 89–101.
[12] P.J. Davis, Circulant Matrices, Wiley, New York, 1979.
[13] L. Eldén, I. Skoglund, Algorithms for the regularization of ill-conditioned least squares problems with tensor product structure, and application to space-variant image restoration, Technical Report LiTH-MAT-R-82-48, Department of Mathematics, Linköping University, Sweden, 1982.
[14] G.H. Golub, M. Heath, G. Wahba, Generalized cross-validation as a method for choosing a good ridge parameter, Technometrics 21 (1979) 215–223.
[15] G.H. Golub, C. Van Loan, Matrix Computations, 3rd ed., Johns Hopkins University Press, Baltimore, MD, 1996.
[16] R.C. Gonzalez, R.E. Woods, Digital Image Processing, Addison-Wesley, Reading, MA, 1993.
[17] C.W. Groetsch, The Theory of Tikhonov Regularization for Fredholm Integral Equations of the First Kind, Pitman, Boston, 1984.
[18] R.J. Hanisch, WF/PC simulation data sets, in: R.J. Hanisch (Ed.), Newsletter of STScI's Image Restoration Project, 1993, 76–77.
[19] M. Hanke, Conjugate Gradient Type Methods for Ill-Posed Problems, Pitman Research Notes in Mathematics, Longman, Harlow, Essex, 1995.
[20] M. Hanke, P.C. Hansen, Regularization methods for large-scale problems, Surv. Math. Ind. 3 (1993) 253–315.
[21] M. Hanke, J.G. Nagy, Restoration of atmospherically blurred images by symmetric indefinite conjugate gradient techniques, Inverse Problems 12 (1996) 157–173.
[22] P.C. Hansen, The truncated SVD as a method for regularization, BIT 27 (1987) 534–553.
[23] P.C. Hansen, Analysis of discrete ill-posed problems by means of the L-curve, SIAM Review 34 (1992) 561–580.
[24] A. Katsaggelos, Digital Image Restoration, Springer, Berlin, 1991.
[25] A.K. Katsaggelos, M.G. Kang, M.R. Banham, Adaptive regularized restoration algorithms applied to HST images, in: R.J. Hanisch, R.L. White (Eds.), The Restoration of HST Images and Spectra II, Space Telescope Science Institute, Baltimore, MD, 1994, pp. 3–13.
[26] J.G. Nagy, Decomposition of block Toeplitz matrices into a sum of Kronecker products with applications in image restoration, Technical Report 96-1, Department of Mathematics, Southern Methodist University, 1996.
[27] J.G. Nagy, R.J. Plemmons, T.C. Torgersen, Iterative image restoration using approximate inverse preconditioning, IEEE Trans. on Image Processing 15 (1996) 1151–1162.
[28] M.K. Ng, R.J. Plemmons, S. Qiao, Regularized iterative blind deconvolution, in: G. Golub, S. Liu, F. Luk, R. Plemmons (Eds.), Proceedings of the HK97 Conference on Scientific Computation, Springer, Berlin, 1997, pp. 10–31.
[29] N.P. Pitsianis, The Kronecker product in approximation and fast transform generation, PhD thesis, Cornell University, Ithaca, NY, 1997.
[30] W.K. Pratt, Vector space formulation of two-dimensional signal processing operations, Computer Graphics and Image Processing 4 (1975) 1–24.
[31] M.C. Roggemann, B. Welsh, Imaging Through Turbulence, CRC Press, Boca Raton, 1996.
[32] S. Thirumalai, High performance algorithms to solve Toeplitz and block Toeplitz matrices, PhD thesis, University of Illinois, Urbana, IL, 1996.
[33] C.F. Van Loan, N.P. Pitsianis, Approximation with Kronecker products, in: M.S. Moonen, G.H. Golub (Eds.), Linear Algebra for Large Scale and Real Time Applications, Kluwer Academic Publishers, Dordrecht, 1993, pp. 293–314.
[34] C. Vogel, M. Oman, A fast, robust algorithm for total variation based reconstruction of noisy blurred images, IEEE Trans. Image Processing 1996 (submitted).

[35] G. Wahba, Practical approximate solutions to linear operator equations when the data are noisy, SIAM J. Numer. Anal. 14 (1977) 651–667.
[36] G.M. Wing, J.D. Zahrt, A Primer on Integral Equations of the First Kind, SIAM, Philadelphia, PA, 1991.